

HCI QUIZ 3 STUDY GUIDE

Slide Set 09: Design Perceptual Organization & Slide Set 10: Usability Evaluations

Exam Scope Warning: Tomorrow's in-class Quiz 3 strictly covers Slide Sets 9 and 10. Expect questions on **Gestalt grouping principles**, visual search constraints, **Formative vs. Summative evaluations**, and **Nielsen's 10 Usability Heuristics** with real-world violations. Review this guide carefully to secure maximum marks.

1. Design Perceptual Organization (Slide Set 09)

Our visual system possesses advanced low-level processing capabilities that influence how we perceive, organize, and scan graphical interfaces. Understanding visual constraints is key to designing high-efficiency user interfaces.

A. Pre-Attentive Processing

Pre-Attentive Processing refers to the rapid, subconscious extraction of visual properties by our low-level sensory system. This happens almost instantaneously (typically under 100 ms) and with high accuracy before conscious focus is established.

- **Single Feature Saliency:** Visual properties such as color (hue), orientation, size, and line width are pre-attentively salient. For instance, detecting a red circle among blue circles is instantaneous.
- **Conjunction Search Bottleneck:** If target detection requires combining multiple features (e.g., finding a red circle among red squares and blue circles), pre-attentive search fails. The user's visual system must switch to a slow, conscious, and *sequential search* across each element.
- **HCI Design Lesson:** Crucial buttons (like warning/danger indicators) should rely on a single unique visual feature to pop out instantly. Over-using multiple color/shape patterns causes visual clutter and destroys visual search efficiency.

B. Gestalt Principles of Visual Grouping

The human mind naturally aggregates visual stimulus into whole structures based on proximity, shape, movement, and alignment. Designers must exploit these natural tendencies to structure forms and content grids.

Principle	Core Scientific Definition	HCI / UI Application & Example
Proximity	Objects positioned closely together are grouped into a single visual unit.	Group related fields (e.g., labels close to inputs). Separate unrelated groups with explicit whitespace.
Similarity	Objects sharing physical properties (color, shape, scale) are grouped.	Format all primary action buttons identically (e.g., green fill). Users expect identical styles to have identical actions.
Continuity	The human eye naturally follows a continuous line or curved path.	Align input forms vertically. Misaligned text fields disrupt the scanning flow and force sequential ocular adjustments.

Closure	The visual system completes incomplete figures or closes open gaps.	Utilize margins and whitespace as active invisible boundaries instead of drawing heavy, distracting boxes or borders.
Figure-Ground	We partition visual scenes into a target (figure) and background (ground).	Make popups and modals clear 'figures' by utilizing distinct drop-shadows and darkening the 'ground' background.
Common Fate	Elements moving together in the same direction/speed are grouped.	In desktop interfaces, dragging a group of items must show them moving in union as a cohesive transparent copy cluster.

2. Usability Engineering & Evaluation Models (Slide Set 10)

Developing usable software requires empirical measurement, structured heuristics, and systematic evaluations. HCI separates usability into rigid measurable attributes and evaluation processes.

A. ISO & Nielsen Usability Framework

The **ISO 9241-11** standard defines usability as the extent to which a product can be used by specified users to achieve specified goals with **effectiveness, efficiency, and satisfaction** in a specified context of use. Jakob Nielsen breaks this down into 5 key metrics:

- **Learnability:** The ease with which novice users can accomplish basic tasks upon their first exposure to the interface.
- **Efficiency:** The high-speed performance achieved by expert users once they have mastered the system.
- **Memorability:** The ease with which returning/intermittent users re-establish proficiency after a period of inactivity.
- **Errors:** The rate, severity, and ease of recovery from mistakes committed by users during system execution.
- **Satisfaction:** The subjective pleasure, comfort, and emotional fulfillment experienced by users during interaction.

B. Formative vs. Summative Evaluation

Evaluations can be classified by their timing and goals during the product's development lifecycle:

Metric / Dimension	Formative Evaluation	Summative Evaluation
Core Timing	Conducted during the design & development phase.	Conducted after the product has been completed or released.
Primary Goal	To diagnose design flaws, identify bugs, and refine the interface.	To assess overall system quality, meet benchmarks, or verify standards.
Data Type Gathered	Mainly qualitative process data (verbal walkthrough comments, user opinions).	Mainly quantitative bottom-line data (time-to-task, error rates).
Core Focus Question	Why did a user struggle or fail to complete a specific task?	What is the exact completion rate and how much did they achieve?

C. Heuristic Evaluation (Nielsen's Method)

A usability inspection method where a small team of experts evaluates an interface against a predefined set of usability principles (the heuristics).

- **The Evaluators Rule:** A single evaluator is highly unreliable and misses roughly 65% of usability issues. Applying **3 to 5 evaluators** is the industry standard. It balances cost and returns, successfully capturing 75-80% of all major usability errors.
- **Severity Ratings:** Used to prioritize fixes during engineering. Measured on a scale of 0 to 4: **0:** No usability problem; **1 (Cosmetic):** Fix only if time permits; **2 (Minor):** Low priority usability flaw; **3 (Major):** High priority,

must-fix usability issue; **4 (Usability Catastrophe)**: Fatal error. System cannot be released without fixing this.

3. Nielsen's Ten Usability Heuristics (Slide Set 10)

These ten principles form the core checklist of heuristic evaluation. Master the scientific definition and standard examples for the quiz.

1. Visibility of System Status

The system must keep users informed about what is going on through appropriate feedback within a reasonable time.

Response Time Constraints: <0.1s perceived as instantaneous (ideal for mouse hover/button clicks); 1s delay is noticed but the user's flow of thought is uninterrupted; 10s is the absolute limit for keeping attention. For tasks longer than 10s, a progress bar and background processing are mandatory.

2. Match Between System and Real World

The system should speak the user's language. Use words, phrases, and concepts familiar to the user rather than internal developer codes or technical jargon. Use real-world metaphors (such as the desktop, folder icons, and the trash can) to align with users' physical mental models.

3. User Control and Freedom

Users often choose system functions by mistake. Provide clearly marked 'emergency exits' to leave unwanted states without an extended dialog. Fully support **Undo**, **Redo**, and **Cancel** operations.

4. Consistency and Standards

Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.

- **Internal Consistency:** Maintain uniform layout grids, color choices, and button locations across all app screens.
- **External Consistency:** Follow global design norms (e.g., placing the shopping cart icon at the top right).

5. Error Prevention

A careful design that prevents errors from occurring in the first place is far superior to writing a great error message.

- **Constraints:** Instead of a text field for date entry (which invites formatting errors), use a calendar popup widget.
- **Reasonableness Checks:** Add validation popups if an entry is unusually large (e.g., ordering 5000 pencils instead of 5).

6. Recognition Rather Than Recall

Minimize the user's memory load by making objects, actions, and options highly visible. The user should not have to memorize information from one part of a dialog to another. Show options, formatting examples (e.g., 'e.g., 0300-1234567'), and dropdowns instead of blank command lines.

7. Flexibility and Efficiency of Use

Provide shortcuts and accelerators (unseen by novice users) to speed up interaction for expert users. Allow users to tailor frequent actions. Examples: keyboard commands (Ctrl+S), custom toolbars, double-clicks, and split menus displaying recently used fonts or files at the top.

8. Aesthetic and Minimalist Design

Dialogs must not contain information that is irrelevant or rarely needed. Every extra unit of information in a dialog competes with the relevant units, diminishing their relative visibility. Keep interfaces clean and decluttered.

9. Help Users Recognize, Diagnose, and Recover from Errors

Error messages must be expressed in plain language (no cryptic system codes like *Error Code 0x80070002*), precisely pinpoint the problem, suggest constructive, helpful solutions, and never make the user feel blamed or foolish.

10. Help and Documentation

Any documentation must be easy to search, focused on the user's task, list concrete steps, and be reasonably compact. Common types of help: tutorials/getting started guides, reminders (tooltips), reference manuals for experts, and wizards.

4. Common Design Violations & Solved Past Paper Questions

This section analyzes legendary design violations (Interface Hall of Shame) from your course materials and solves the exact conceptual questions from the UCP Midterm past papers.

A. Classic Interface Hall of Shame Breakdowns

- **Cryptic Error Messages (e.g. Visual Basic Blank Dialogs):** Violates Heuristics 1 & 9. Popups with only an 'OK' button and no text or codes like 'Error &H80004005: Unspecified error' fail to keep the user informed or suggest recovery steps.
- **Scrollbar for Award Style Selection:** Violates Heuristics 2 & 5. Forcing a scrollbar to slide horizontally to select award styles (instead of showing a standard vertical list of names) violates real-world mental mapping and causes high error rates.
- **Cluttered Layouts (e.g. HavenWorks):** Violates Heuristic 8. Overloading a page with multiple fonts, unaligned columns, and saturated color cards fails pre-attentive visual processing, forcing the user into slow sequential searches.

B. Solved UCP Midterm Exam Questions

Q1: Explain why context is extremely important when evaluating usability.

Solution: Usability is not a static or isolated property of an interface. As defined by ISO 9241, usability is heavily dependent on 'specified users, specified goals, and a **specified context of use**.' For instance, a complex, desktop form-filling screen might have excellent usability in a quiet office environment, but it becomes completely unusable in a high-stress medical ER room where clinicians are moving fast, wearing gloves, and dealing with noise. Usability evaluations must always mimic the actual environmental and physical constraints of the real-world context.

Q2: An elderly person still uses a basic keypad phone instead of a smartphone. Classify this user type and explain your reasoning.

Solution: This individual is classified as an **Extreme User** in terms of technology adoption and accessibility. In user needfinding, extreme users are highly useful because their physical/cognitive limitations or behaviors are highly amplified. By designing solutions that accommodate an extreme user (e.g., high-contrast text, clear tactile physical keys, simple navigation), designers build interfaces with low cognitive load and excellent affordances, which ultimately improves the usability for the entire typical user class.

Q3: A persona named 'Ali' is a first-year student with high exam anxiety. Suggest design features to satisfy this persona.

Solution: To accommodate Ali's anxiety and reduce cognitive stress:

1. *Error Prevention:* Implement visible constraints (e.g., warning prompt: 'Are you sure you want to exit?') and disable the submit button after one click to prevent double-submissions.
2. *Visibility of System Status:* Show a persistent, clear remaining-time clock and immediate 'Answer Saved' feedback on every input.
3. *User Control:* Provide an easy way to bookmark/flag questions to return to later, and full Undo/Redo control.

Q4: Evaluate a booking design that uses Red for 'Keep' and Green for 'Delete' buttons.

Solution: This severely violates ****Match Between System and Real World**** (metaphors/conventions) and ****Consistency and Standards****. In physical world conventions, red universally signifies 'stop/danger/warning' (appropriate for destructive actions like Delete) while green signifies 'go/safety/confirm' (appropriate for Keep). Reversing this mapping creates a critical cognitive trap, leading to accidental data deletions and high user error rates.